

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

Ans. Common BIOS / POST Error Messages

No.	Error Message	Usually Points To
1.	CMOS Checksum Error	CMOS battery or BIOS settings problem.
2.	Keyboard Error	Keyboard is not connected properly or is faulty.
3.	No Boot Device Found	HDD/SSD is not detected or the boot drive is missing.
4.	Memory Error	RAM may be loose, damaged, or not detected.
5.	CPU Fan Error	CPU cooling fan may not be connected or may not be working.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Ans. 3 Beeps on Boot – AMI BIOS

For a traditional **AMI BIOS**, **3 short beeps usually mean a memory (RAM) error**—the BIOS found a problem while testing the first part of the system memory.

Two easy troubleshooting steps:

1. **Turn off the PC and reseat the RAM** — remove it and put it back firmly in the slot.
2. **Test one RAM stick at a time** — if the PC starts with one stick, the other RAM stick may be faulty.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

Ans.



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Think about the CMOS battery and BIOS settings reset.

Ans.



What does it mean?

This message means the BIOS settings stored in the CMOS memory are not correct or have been lost. The system detected a checksum error in the CMOS, so it loaded the default settings instead of your custom settings.



How to fix it

- 1 Turn off the PC and unplug the power cable.
- 2 Locate the CMOS battery on the motherboard (it's a small round battery).
- 3 Remove the battery for a few minutes (or replace it if it's old or dead).
- 4 Put the battery back, plug in the power cable, and start the PC.
- 5 Enter BIOS setup (usually by pressing DEL or F2) and set the correct date, time and other settings.
- 6 Save and exit.